Kilnbright

Executive Summary

Kilnbright
A Smart Kiln Controller for Classroom Learning

Problem

In middle schools across America, art teachers have a problem with kilns. Kilns is very hot, up to over two thousand degrees, and are danger if the teacher do not know how to use it correct. Many teachers are not trained for firing ceramics and they are afraid of the kiln so some schools dont even use there kiln at all and it sit in the corner. This is a big problem because ceramics is a important part of art education and students loses the chance to make things with there hands. Also the old kiln controllers is hard to understand with many buttons and numbers and codes that the teacher must learn. When the teacher make mistake the pottery can break or explode and this waste time and money and the students feels bad. Also the students do not learn nothing about the science of heat during firing, which is a missed opportunity in STEM education. We think this problem is very big and it need to be solved.

Solution

Kilnbright is the solution. It is a low cost digital kiln controller that is classroom safe. The teacher just plug it into the kiln and choose a preset program like bisque or glaze and push the button. Then Kilnbright do everything automatic and the teacher dont need to understand the temperatures. It have a safety feature that shut off the kiln if it get too hot. It also have a color screen who show the students what is happening inside the kiln and explain the science of heat and glaze chemistry in a fun way. So Kilnbright is both a controller and a education tool. We believe this is a innovative and disruptive solution that will change how ceramics is taught. Our controller is easy, safe, smart, and fun, and it is the first to combine all of these things together in one product for schools.

Market

The market for Kilnbright is very large. There is thousands of middle schools in the United States and each one have a art teacher and many have a kiln. Education is a huge market worth billions of dollars and technology in education is growing every year. If we get only one percent of the market we will make a lot of money. Also there is private art studios, summer camps, community centers, and hobby potters who might want our product too, so the market is even bigger. We think the total market is over one billion dollars. Schools are always looking for new STEM products because of the focus on STEM in education and the government give funding for it. Teachers also buy things themselves with there own money for the classroom. The market is growing and our timing is perfect because everybody is talking about STEM and hands on learning right now.

Business Model

We will make money by selling the Kilnbright controller to schools and teachers. The price will be about two hundred dollars which is low cost compared to other controllers. We will also sell a subscription for lesson content and new firing programs, maybe around ten dollars a month, but we have not decided yet the exact price. Later we might also sell accessories and other products and maybe advertising or partnerships. We will sell online through our website and also to school districts directly. We think many schools will buy it because it is cheap and the teachers will tell other teachers about it so the sales will grow fast by word of mouth. The hardware have a good margin and the subscription will make recurring revenue so the business is scalable. We will figure out the details of the pricing after we talk to more customers.

Competition

There is some competitors who sell kiln controllers but they are not like us. The existing controllers is expensive, complicated, and made for professional potters not for classrooms. They dont have education features and are not safe for kids. Some companies also make kilns with built in controllers but this is more expensive and school must buy whole new kiln. Because of this we think we have no real competition in the classroom market. Our competitive advantage is that we are the only one who is focus on students and teachers and make it easy and educational. Big companies could maybe copy us but we will be first to market and have the brand. Also we have a website and we are working on getting patent protection for our idea, which will protect us from competition.

Team

Dashiell Brownleigh is the CEO and is a junior studying business at our university. He is passionate about education and entrepreneurship and he did a internship once. He lead the team and do the strategy and sales.

Marisol Quentara is the CTO and is a senior studying electrical engineering at our university. She build the prototype and know about electronics and sensors and programming. She is very talented.

Tobias Wrenfield is head of product and a sophomore studying art education at our university. He have experience with ceramics and understand the teachers needs because he took a pottery class and work in the art department.

We are a strong team with business, technology, and art skills together and we work very hard. We dont have yet a advisor but we are looking for mentors.

Milestones and Financials

So far we have defined our core business concept and we have build a working prototype of the Kilnbright controller. The prototype can control a small kiln and run a preset program and show the screen. We also raised some funding, about eight thousand dollars, from family and friends and a small grant from our university. We have a website also. We have not formed a legal entity yet but we will do it soon when we need it. We have not sold anything yet and so we have no revenue at the moment. We have talked to some teachers who said they like the idea, but we have not done formal customer research yet. For the next year we plan to improve the prototype, make a bigger version, test in some schools, and start selling. We think we can have revenue of one million dollars in three years and be profitable soon after, but we dont have a detailed financial model yet. The cost to make each controller is maybe around eighty dollars but this is a estimate and could change.

What We Are Asking For

We are asking for the prize money from this competition to help us grow Kilnbright. We would use the money to build more prototypes, pay for safety certification, hire a manufacturer, and do marketing. We are also asking for mentorship and connections to investors and schools and any help with legal and business advice. We believe that Kilnbright can be a big success and make a real impact in education and we are excited to join this competition. Thank you for considering our application and we hope to be selected so we can take Kilnbright to the next level and make ceramics safe and fun for every classroom in America.